Astronomy - Graduate Level Assessment 9 Total Marks: 40 Answer all questions.

# Multiple Choice (10 marks)

1. Mars might be a place for future human explorations. However humans can not breathe on the surface of Mars because the atmosphere consists mostly of ...
   (a) Nitrogen
   (b) Argon
   (c) Methane
   (d) CO 2
2. What would weigh the most on the moon?
   (a) A kilogram of feathers
   (b) Five pounds of bricks as measured on Earth
   (c) Five kilograms of feathers
   (d) A kilogram of bricks
3. You've made a scientific theory that there is an attractive force between all objects. When will your theory be proven to be correct?
   (a) The first time you drop a bowling ball and it falls to the ground proving your hypothesis.
   (b) After you've repeated your experiment many times.
   (c) You can never prove your theory to be correct only "yet to be proven wrong".
   (d) When you and many others have tested the hypothesis.
4. One astronomical unit (AU) is equal to approximately ...
   (a) 130 million km
   (b) 150 million km
   (c) 170 million km
   (d) 190 million km
5. The visible part of the electromagnetic spectrum is between ...
   (a) 240 to 680 nm.
   (b) 360 to 620 nm.
   (c) 380 to 740 nm.
   (d) 420 to 810 nm.

# True / False (10 marks)

*Answer True or False*

6. True or False: The Tunguska event in Siberia is associated with the mass extinction of dinosaurs 65 million years ago.

7. True or False: The United States has sent landers to Venus that successfully operated on the planet's surface and transmitted data back to Earth.

8. True or False: Titan's atmosphere is predominantly composed of nitrogen and oxygen, similar to Earth's atmosphere.

9. True or False: The solar radiation flux on Mercury's surface at perihelion is approximately 9/4 times greater than at aphelion.

10. True or False: A day on the Moon, measured from one sunrise to the next, lasts approximately 29 Earth days.

# Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the essential characteristics that formation theories of our solar system must address, and critically evaluate why the presence of life on Earth is not included in these explanations.

12. Discuss the methods used to determine the distance of the rock, identified as "bigfoot" on Mars, based on its angular size and physical dimensions. Include calculations and implications for rover observations.

*End of Paper*